

MarketMind AI

Small Business Sales Intelligence Platform — Research Dashboard Layouts

Project Objective: Build a full-stack AI-powered sales intelligence platform to help businesses monitor performance, track inventory, manage invoices, and generate predictive insights using machine learning.

Dashboard Concept: Role-Based UI layout tailored for Business Owners, Store Managers, Sales Executives, and System Administrators.

Core Tech Stack: FastAPI (Backend) | ReactJS / Streamlit (Frontend) | PostgreSQL / SQLite | Scikit-Learn & Prophet (AI Engine)

1. Business Owner Dashboard View (Executive Strategy)

Primary Focus: High-level strategic decisions, sales forecasting, and churn analysis.

Total Revenue Today \$12,450	Predicted Growth (Q3) +18.4%	At-Risk Customers 14 Users
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MAIN PANEL: Sales Forecasting Engine

- **Algorithm:** Prophet / XGBoost Regressor
- **Visual:** Time-series line chart with solid historical data vs shaded prediction band.
- **Controls:** Date range selector, category filter, seasonal trend toggles.

SECONDARY PANEL: Churn Prediction

- **Algorithm:** Random Forest / XGBoost
- **Visual:** List of high-value customers at risk of inactivity.
- **Output:** Risk probability score + retention recommendations.

2. Store Manager Dashboard View (Operations & Inventory)

Primary Focus: Daily inventory management, stock alerts, and anomaly detection.

Total Inventory Value \$84,200	Reorder Alerts 5 Items	Inventory Anomalies 2 Flagged
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MAIN PANEL: Stock Tracking & Warehouse Matrix

- Progress bar indicators for stock levels (Green: OK, Orange: Low, Red: Critical).
- Instant action button for threshold-based reordering alerts.
- Multi-warehouse inventory logs.

SECONDARY PANEL: Anomaly Detection System

- **Algorithm:** Isolation Forest / Statistical Outlier Detection
- Highlights unusual sales transactions and inventory loss.
- Real-time alert generation for suspicious activities.

3. Sales Executive & Admin Dashboard Views

Sales Executive View (Transactions & POS)

- **Sales Data Upload:** Fast CSV import and transaction storage.
- **Invoice Management:** Create, track, and monitor payment statuses.
- **Product Recommendation Engine:** Collaborative Filtering & Association Rules for real-time cross-sell/upsell suggestions.

System Administrator View (Platform Management)

- **User & Role Management:** Enforce Role-Based Access Control (RBAC).
- **AI Module Configuration:** Monitor ML models, view evaluation metrics (MAE, RMSE, Precision, F1-Score).
- **System Security:** Audit logging, API rate limiting, database status.